

UNIT-5

NETWORK PLANNING AND IMPLEMENTATION

5.1 NETWORK NEEDS

A computer network is a collection of interconnected computers, servers, and other devices that can communicate with each other, either wirelessly or through physical connections like cables or fiber optics. These interconnected devices can share resources such as files, printers and internet connections, and they can also facilitate communication through email, messaging, and other means. Computer networks can vary in size and complexity, ranging from small local networks within a home or office to large-scale global networks like the Internet.

Networks serve various purposes and fulfill multiple needs for individuals, businesses and organizations. Some of the key needs of a network include:

- 1) Communication:** Networks enable communication between individuals, devices and systems regardless of their physical location. This includes email, instant messaging, voice and video calls and data sharing.
- 2) Resource Sharing:** Networks allow for the sharing of hardware resources such as printers, scanners and storage devices, as well as software resources like applications and databases.
- 3) Data Transfer and Access:** Networks facilitate the transfer of data between devices and provide access to shared data repositories, allowing users to retrieve, store and manipulate information.
- 4) Collaboration:** Networks support collaboration among users by enabling them to work together on projects, share documents and coordinate tasks in real-time, regardless of their geographical location.
- 5) Remote Access:** Networks enable remote access to resources and services, allowing users to connect to their organization's network from external locations securely.
- 6) Internet Access:** Networks provide connectivity to the Internet, allowing users to access a vast array of online resources, services and information.
- 7) Centralized Management:** Networks allow for centralized management of resources, user accounts, security policies and network configurations, streamlining administration and maintenance tasks.
- 8) Backup and Disaster Recovery:** Networks facilitate backup and replication of data to remote locations, ensuring data integrity and enabling disaster recovery in case of hardware failures, natural disasters or cyberattacks.
- 9) Scalability:** Networks can be scaled to accommodate the growth of an organization by adding additional devices, expanding infrastructure capacity, and adapting to changing requirements.
- 10) Security:** Networks provide security mechanisms such as firewalls, encryption, access controls and intrusion detection systems to protect against unauthorized access, data breaches and cyber threats.

5.2 INSTALL AND CONFIGURE WINDOWS SERVER

Windows Server 2012 R2 Hardware Requirements:

As with previous Windows versions, your hardware must meet certain requirements for Windows Server 2012 R2 to function properly. First of all, Windows Server 2012 R2 requires a 64-bit processor.

Component	Minimum Requirement	Microsoft Recommended
Processor	1.4 GHz	2 GHz or faster
Memory	512 MB RAM	2 GB RAM or greater
Available Disk Space	32 GB	40 GB or greater
Optical Drive	DVD-ROM drive	DVD-ROM drive
Display	Super VGA (800x600) monitor	XGA (1024x768) monitor

In addition, you must have the usual I/O peripherals, including a keyboard and mouse or compatible pointing device and a wired or wireless network interface card (NIC). If you can connect to a network location on which you have copied the contents of the Windows Server 2012 R2 DVD-ROM, you are not required to have a DVD-ROM drive on your computer.

Microsoft recommends that you also perform the following actions before installing Windows Server 2012 R2:

- 1) Disconnect uninterruptible power supply (UPS) devices
- 2) Back up data
- 3) Disable antivirus software
- 4) Provide mass storage drivers if needed
- 5) Note that Windows Firewall is on by default
- 6) Prepare your Active Directory environment for Windows Server 2012 R2

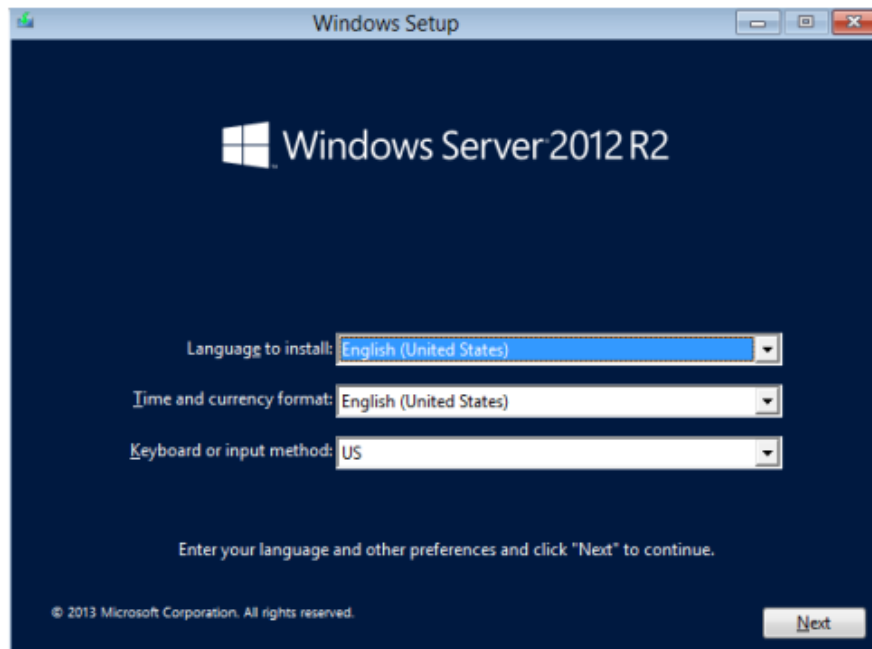
Steps to install Windows Server 2012:

1. Insert the Windows Server 2012 R2 DVD-ROM and turn on your computer. You should see a message informing you that Windows is copying temporary files; if not, you should access the BIOS setup program included with your computer and modify the boot sequence so that the computer boots from the DVD.

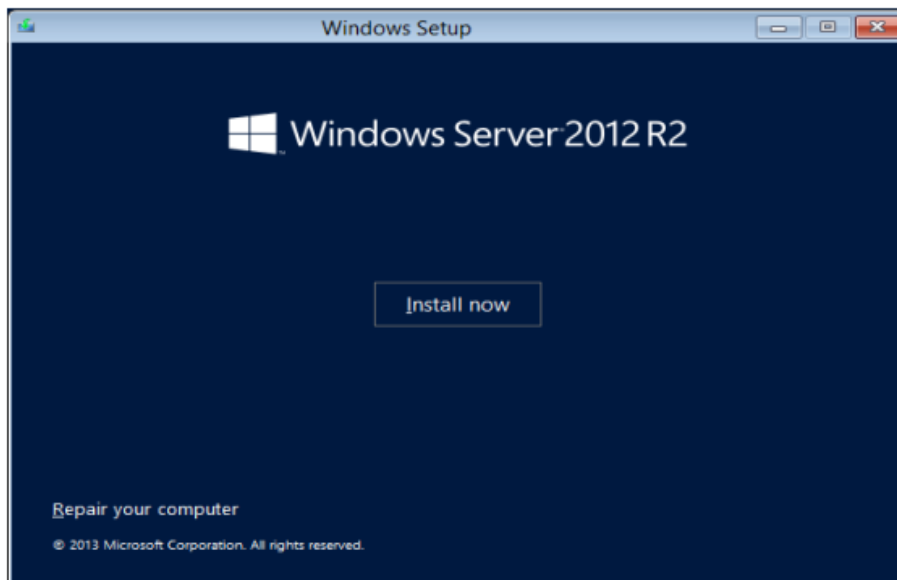
2. Once booted, you should see Windows loading the necessary setup files.



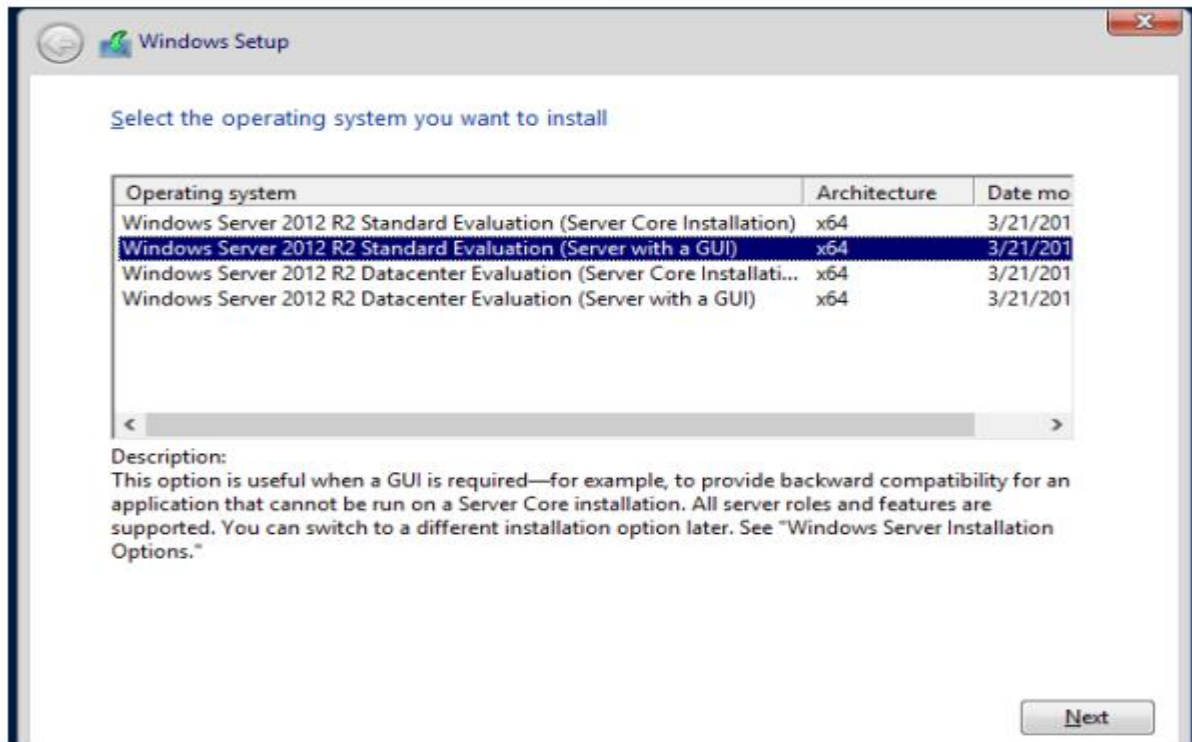
3. In the Windows Setup dialog, set the language, time and currency format, keyboard and input method then press Next.



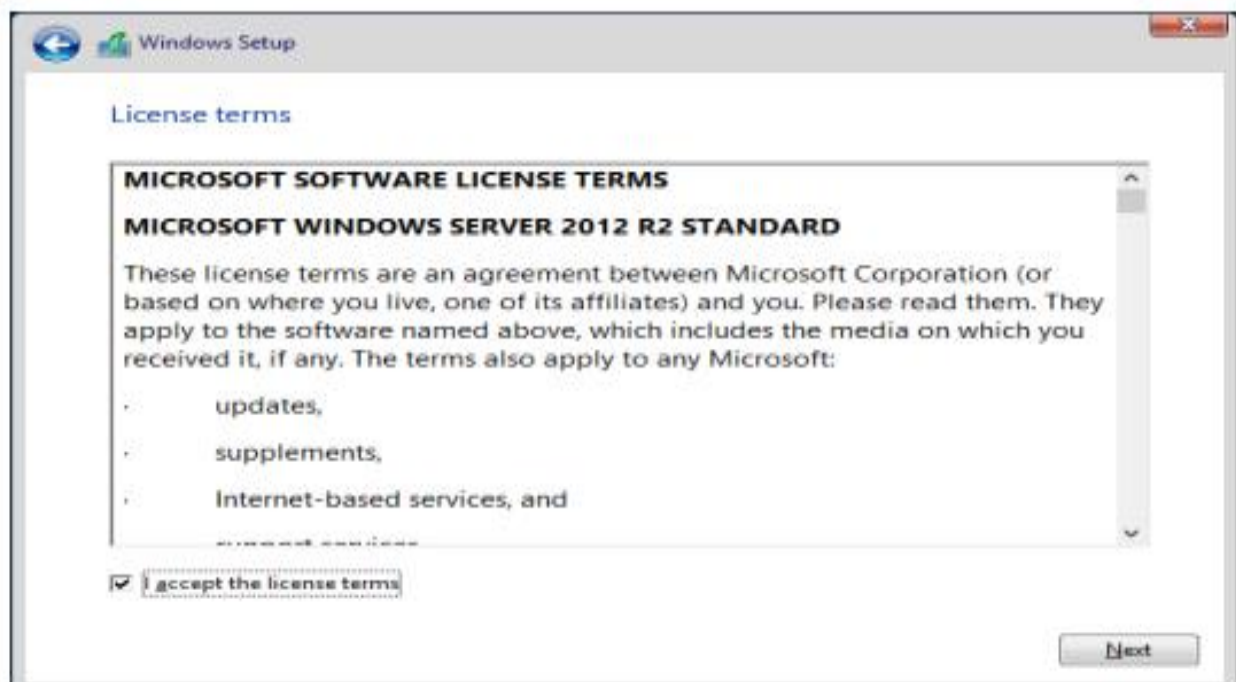
4. Click on Install now



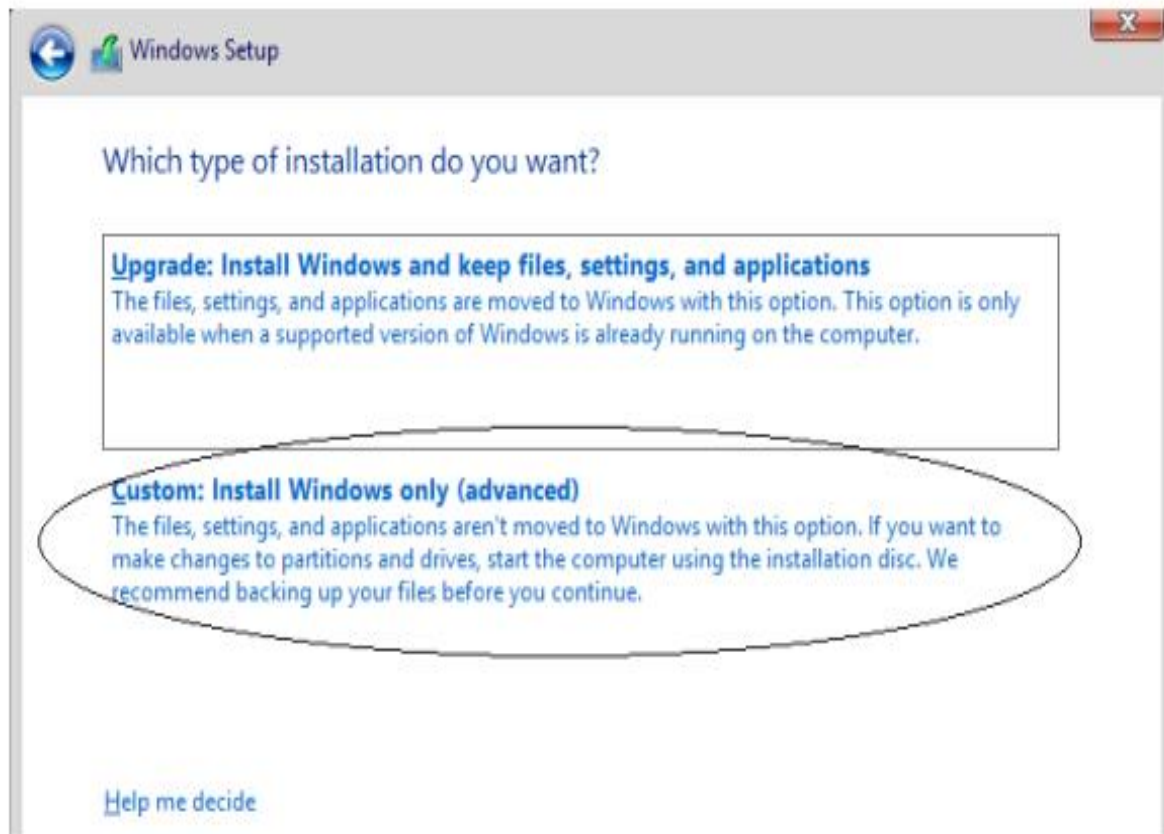
5. Select the operating system that you wish to install. Regardless if you install the Server 2012 R2 Datacenter or the 2012 R2 Standard version, you want the installing package that includes the GUI. Otherwise, you end up with just the Server 2012 Core and no GUI.



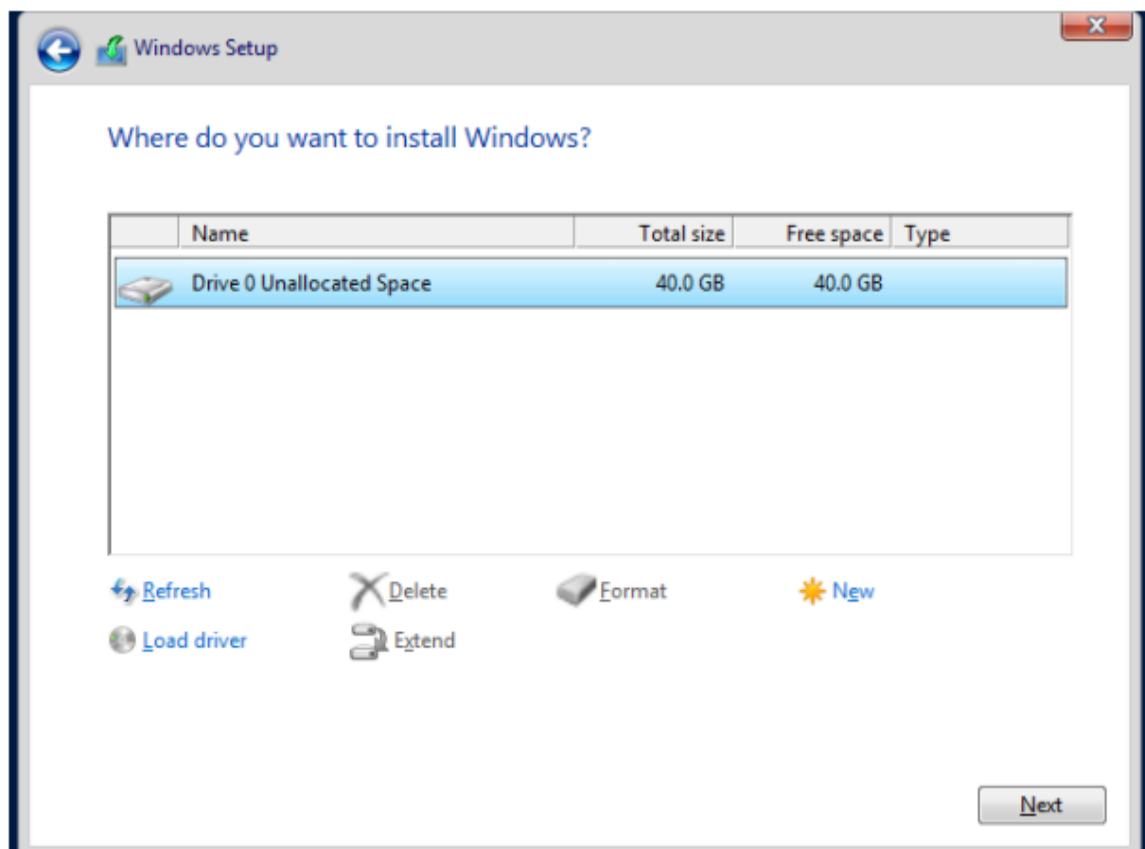
6. Accept the license agreement.



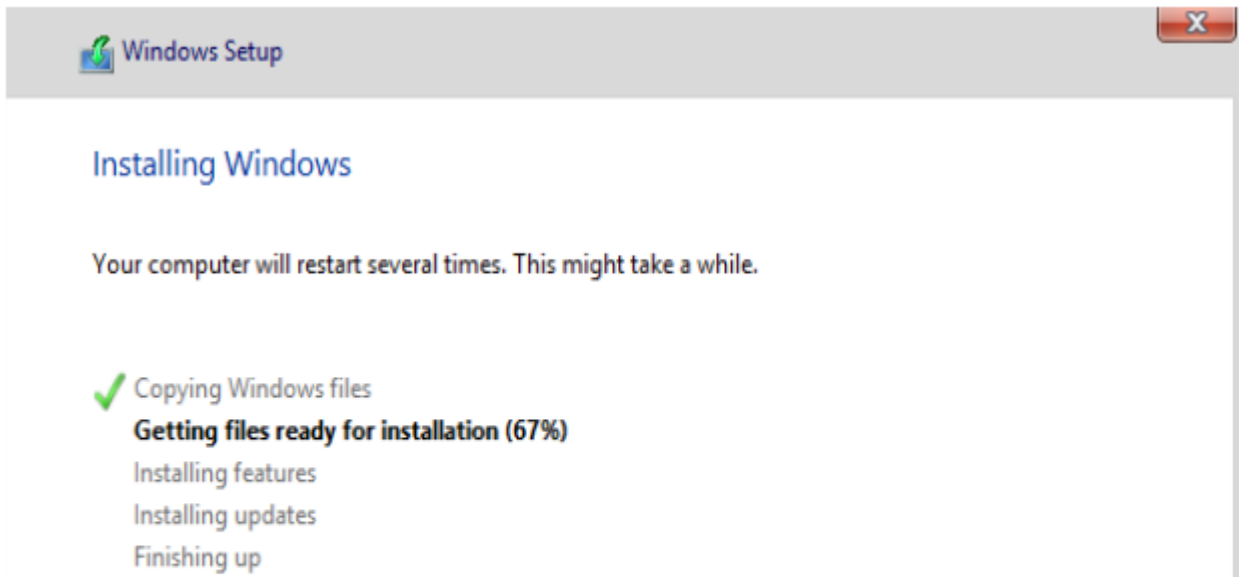
7. Select Custom Install Windows only (advanced).



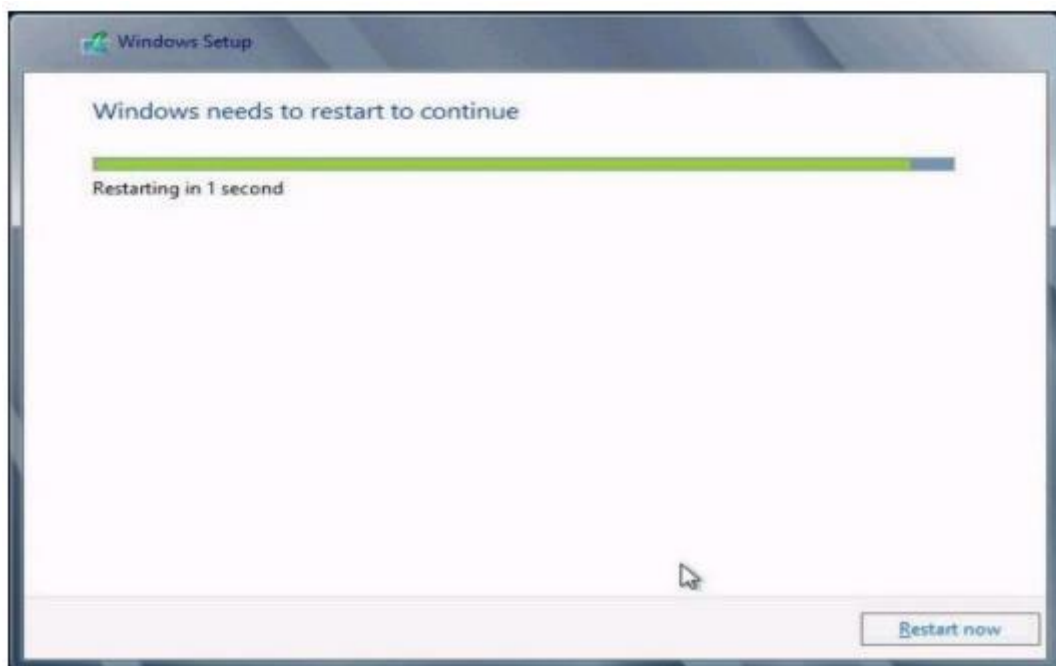
8. Click Next.



9. Server 2012 R2 begins the file copy process.



10. The server will restart automatically and installation will continue further.



11. Before Server 2012 will allow you to log in, you must configure the password for the administrator account.
12. Click Finish. Welcome to Windows Server 2012 R2!

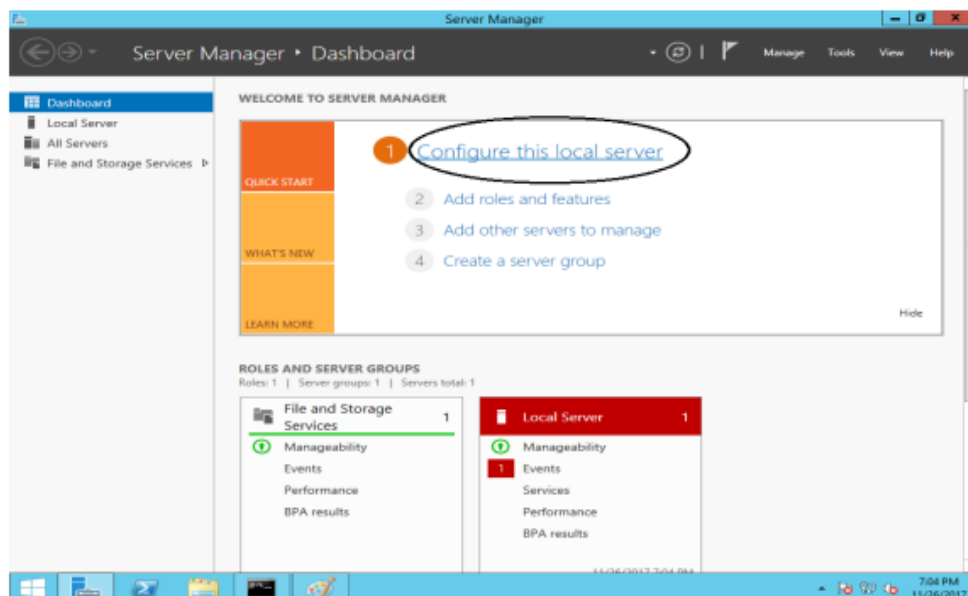


The steps to Change the Server Name are:

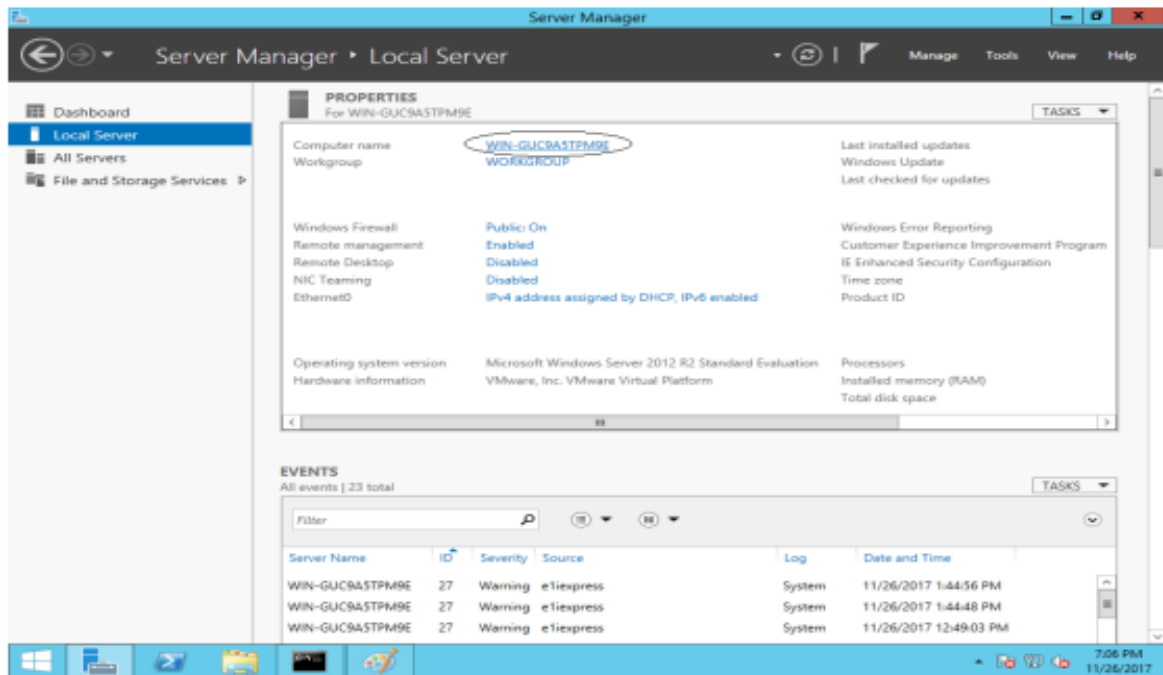
1. Open Server manager



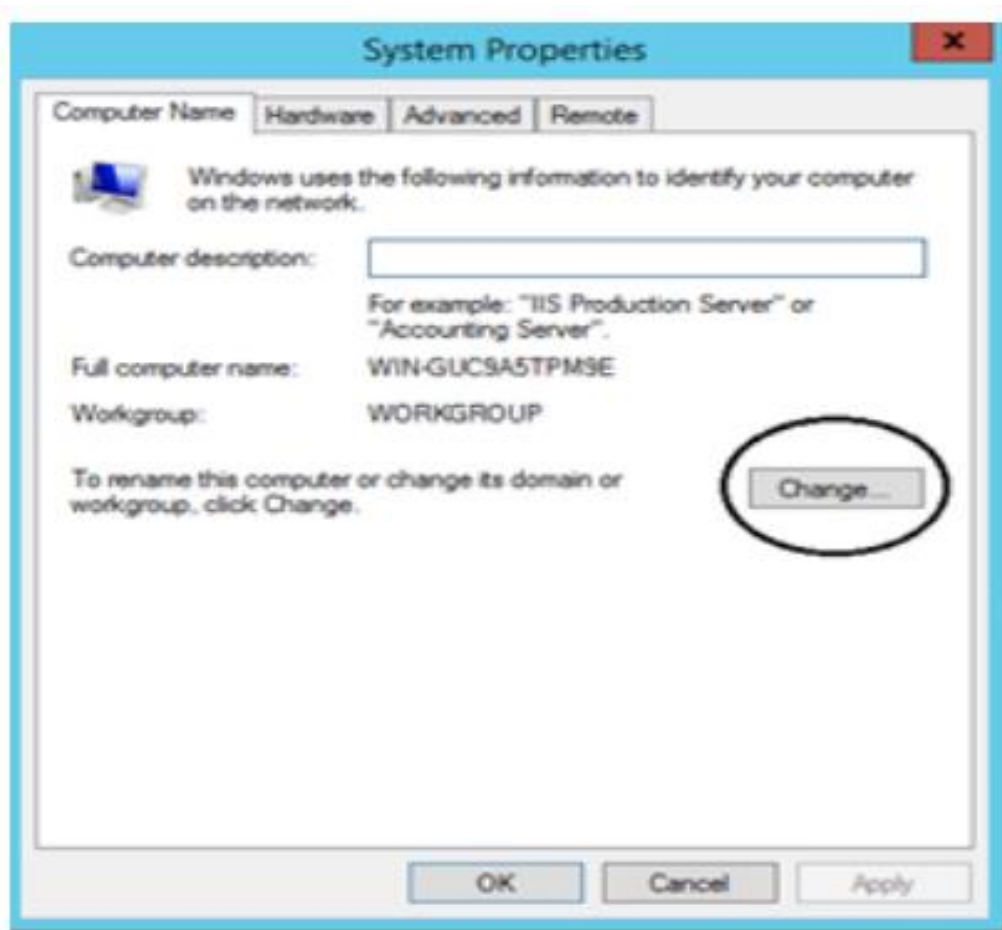
2. Click on Configure this local server



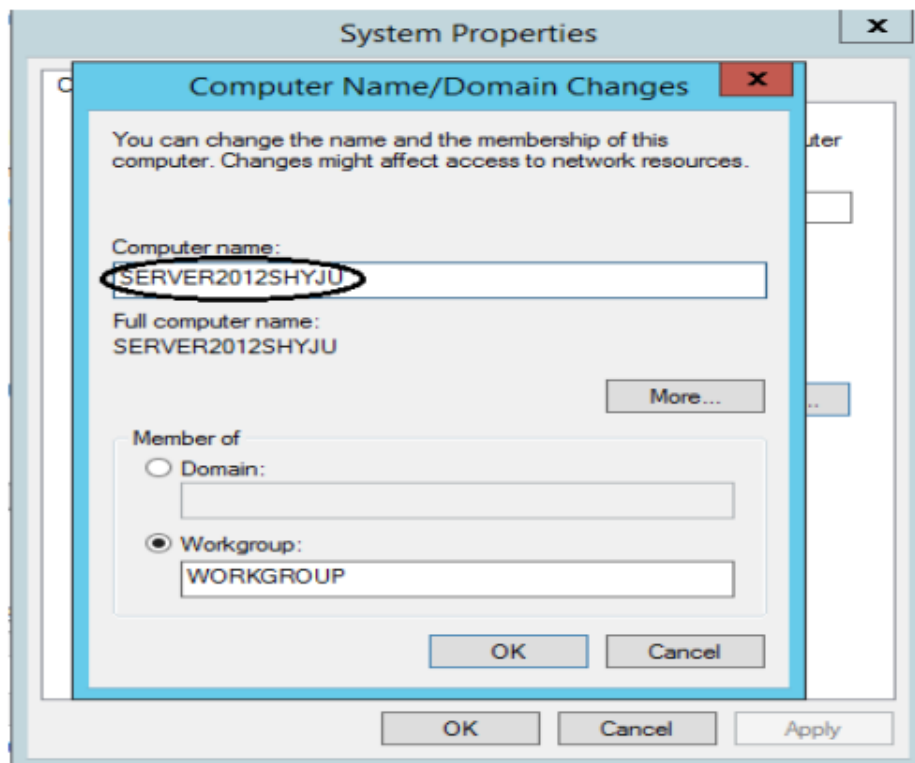
- Click on default Computer name which was given while server 2012 was installed.



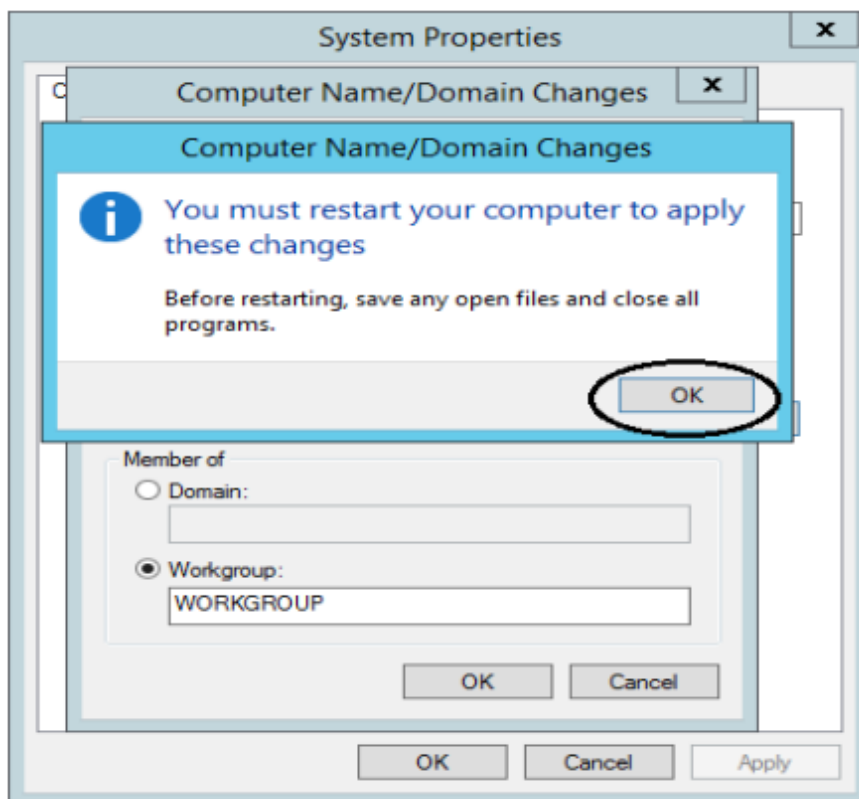
- Click Change in System Properties window



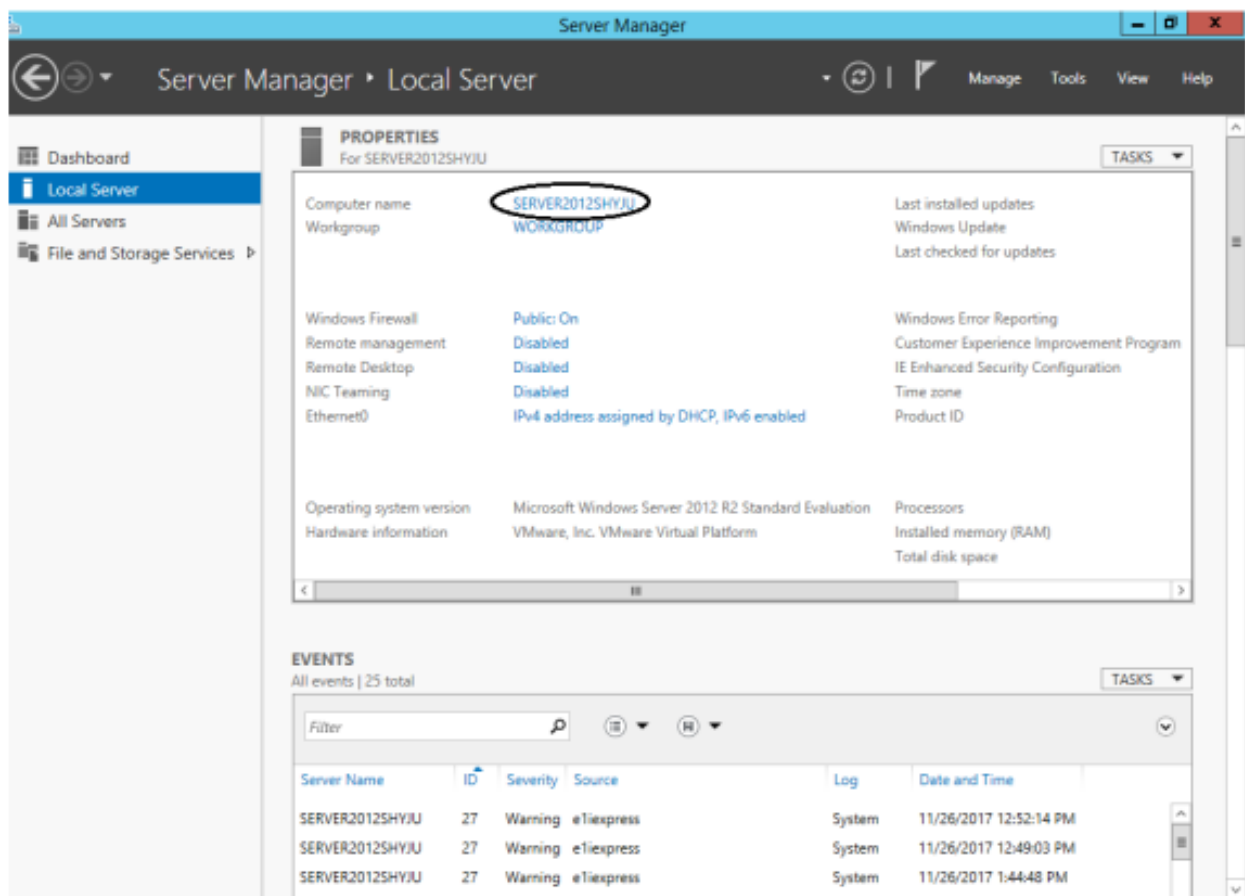
5. Give a new Computer name which is easy to remember.



6. Click OK and Restart your computer.



7. After Restarting your computer server name will be changed



5.3 STEPS TO CREATE DOMAIN CONTROLLER

Set a static IP address

To start, you'll want to ensure your domain controller has a static IP address (that isn't in a DHCP range either) to ensure it cannot change automatically.

Install the Role

From the Server Manager click Manage > Add Roles and Features this will open up the Add Roles and Features Wizard.

Once open, click next until you reach the Server Roles section.

Once there click the check box next to Active Directory Domain Services.

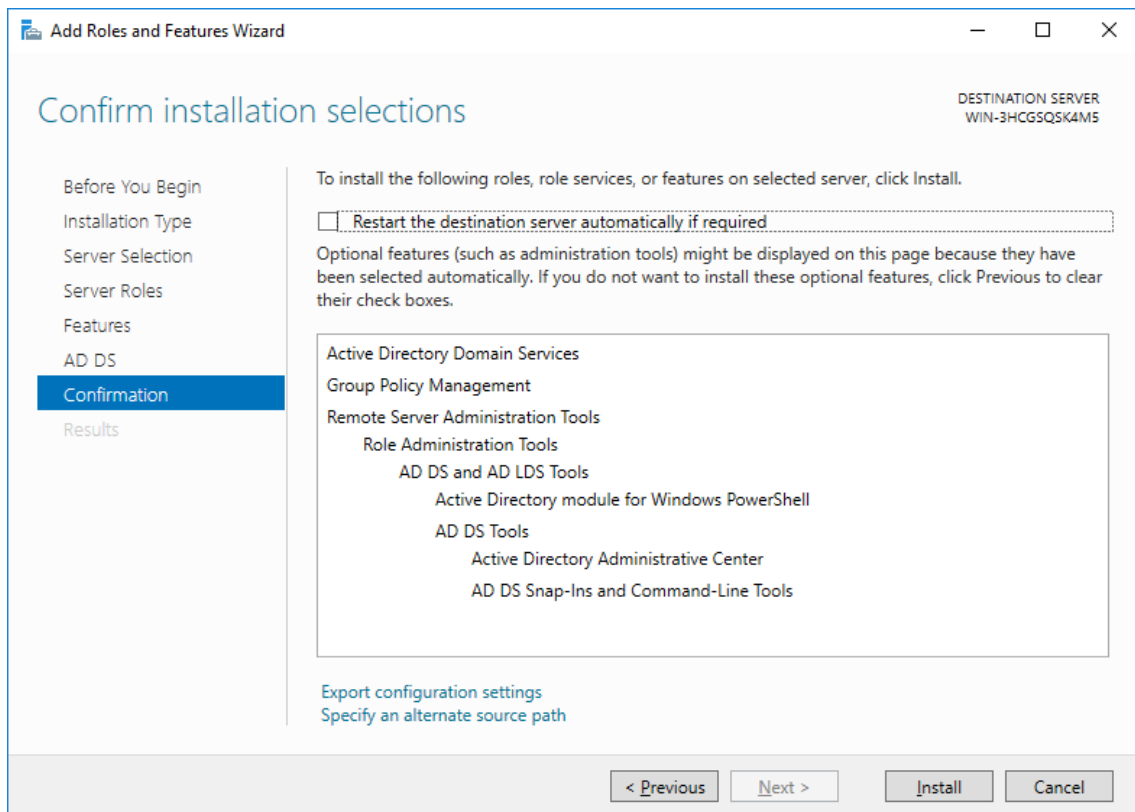
Another window should open asking if you want to install Active Directory Administrative Center and the AD DS Snap-in and Command Line Tools.

We recommend you accept this and click the Add Features button with the Include Management tools (if applicable) box checked as well.

That will be the only role to include this time so you can click next and you can also click next through the features section as well.

Once you reach the confirmation section it will display the list of roles and features to be installed; it should match the image below.

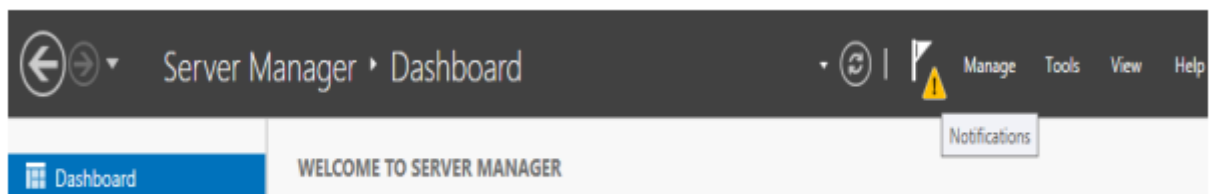
Click install and now we wait!



Promoting Server 2012 R2 to a Domain Controller

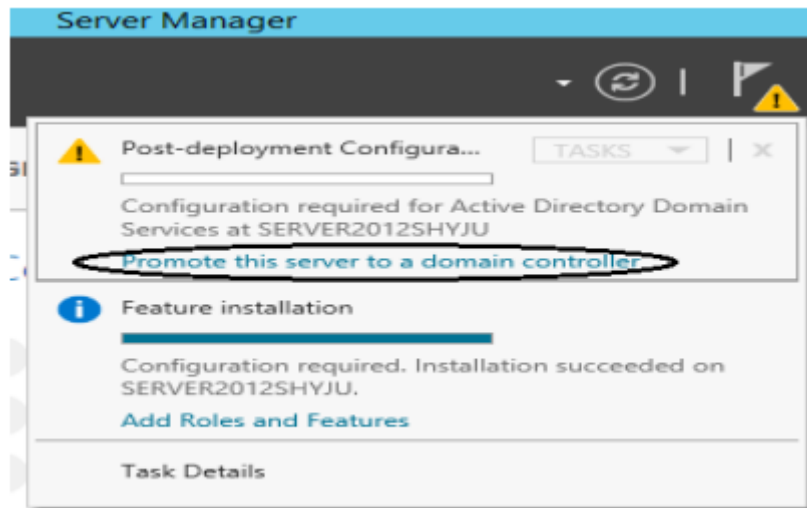
Once the Active Directory Domain Services role has been installed, we can promote our server to a Domain Controller by following the below steps.

1. When we return to Server Manager, there is an information message waiting for us. This is the second part of the role installation that converts the server to a domain controller.

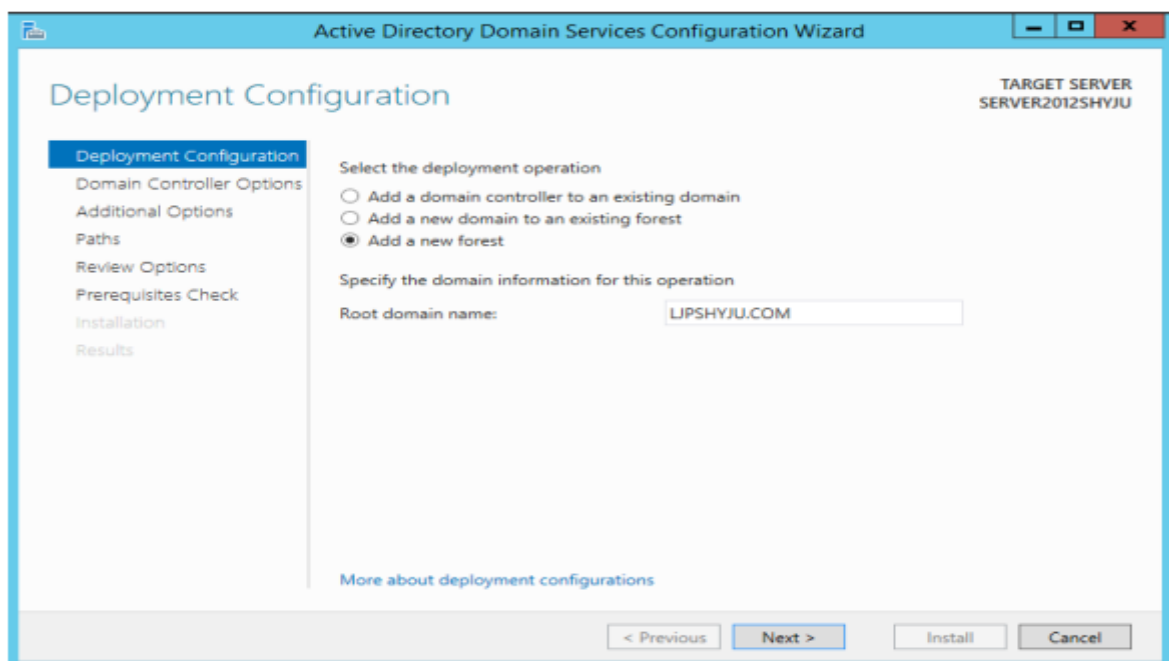


2. Click on Notification you will see the option "Promote this server to a domain controller." Click on this link for role promotion.

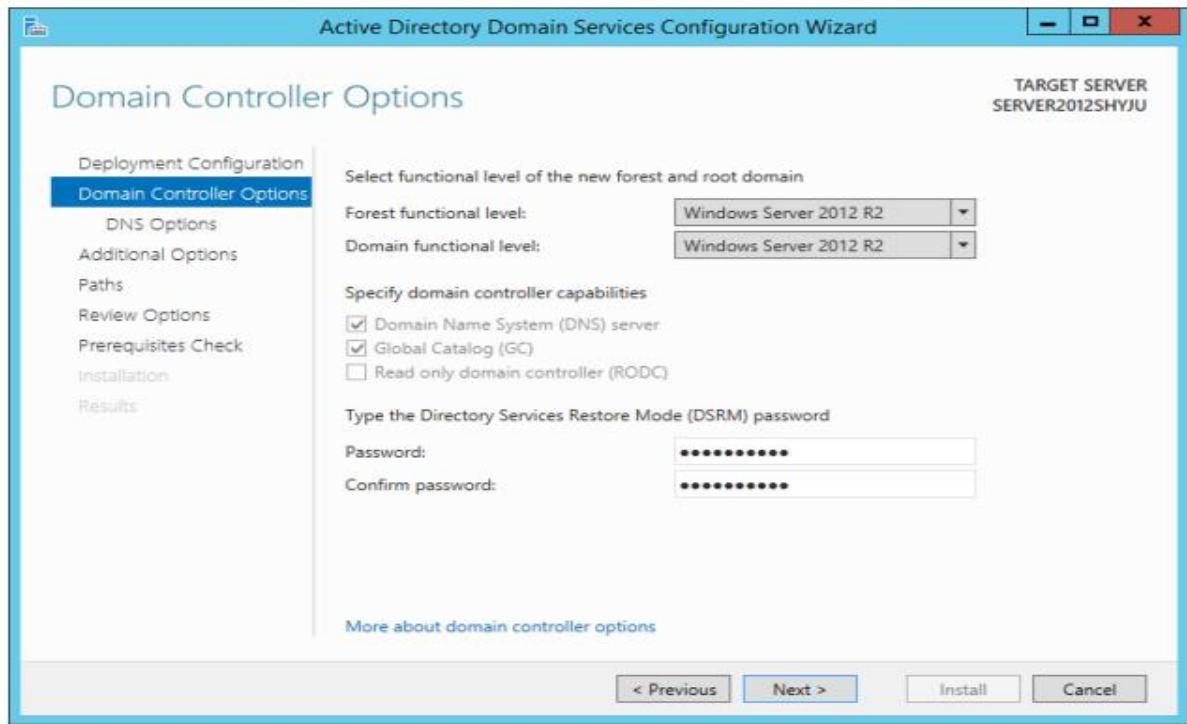
If we click on the message, "Promote this server to a domain controller" the second wizard will start.



3. Click on the “Add a new forest” radio button and enter the name of the new root domain. Click Next

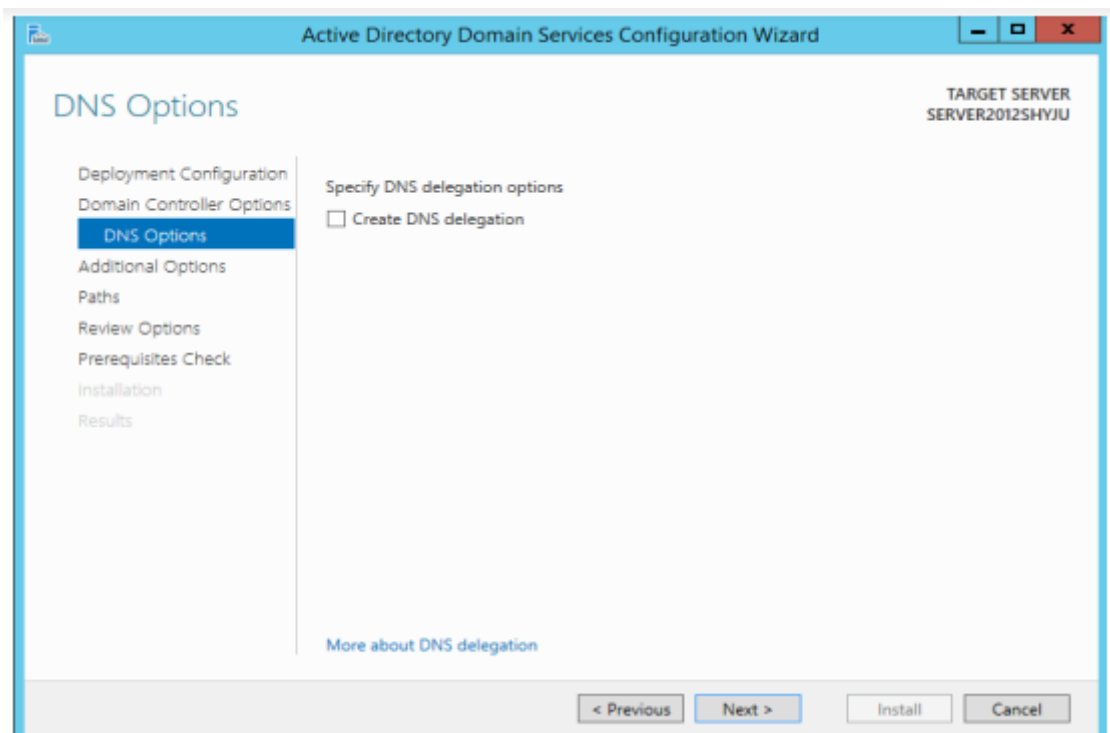


4. If we did not have DNS installed, the option to install DNS would be provided. Check the box to install DNS as it is a requirement for Active Directory. If it is grayed out, you already have it installed and running. Leave the function for Server 2012 R2 as is. If we had server 2008 running on the network, we downgrade the function to accommodate the need for a backward capability. Type the Directory Services Restore Mode (DSRM) password in the Domain Controller Options window.



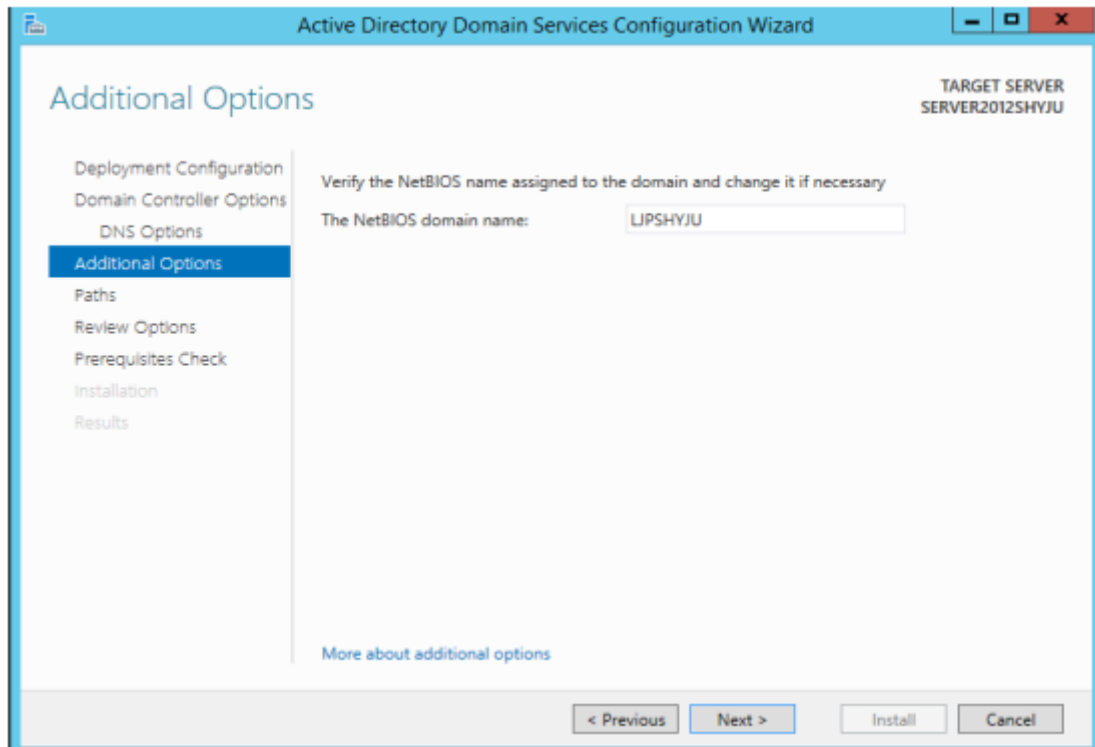
The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo, the text 'Active Directory Domain Services Configuration Wizard', and standard window controls. The main window has a blue header with the title 'Domain Controller Options' and a 'TARGET SERVER' label with the value 'SERVER2012SHYJU'. On the left is a navigation pane with links: 'Deployment Configuration', 'Domain Controller Options' (highlighted), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is titled 'Select functional level of the new forest and root domain'. It contains two dropdown menus: 'Forest functional level:' and 'Domain functional level:', both set to 'Windows Server 2012 R2'. Below these is a section 'Specify domain controller capabilities' with three checkboxes: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked). Further down is a section 'Type the Directory Services Restore Mode (DSRM) password' with two password fields labeled 'Password:' and 'Confirm password:', both containing masked characters. At the bottom of the main area is a link 'More about domain controller options'. The footer contains four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

5. Uncheck the box DNS delegation if authoritative zone is available. Click next.

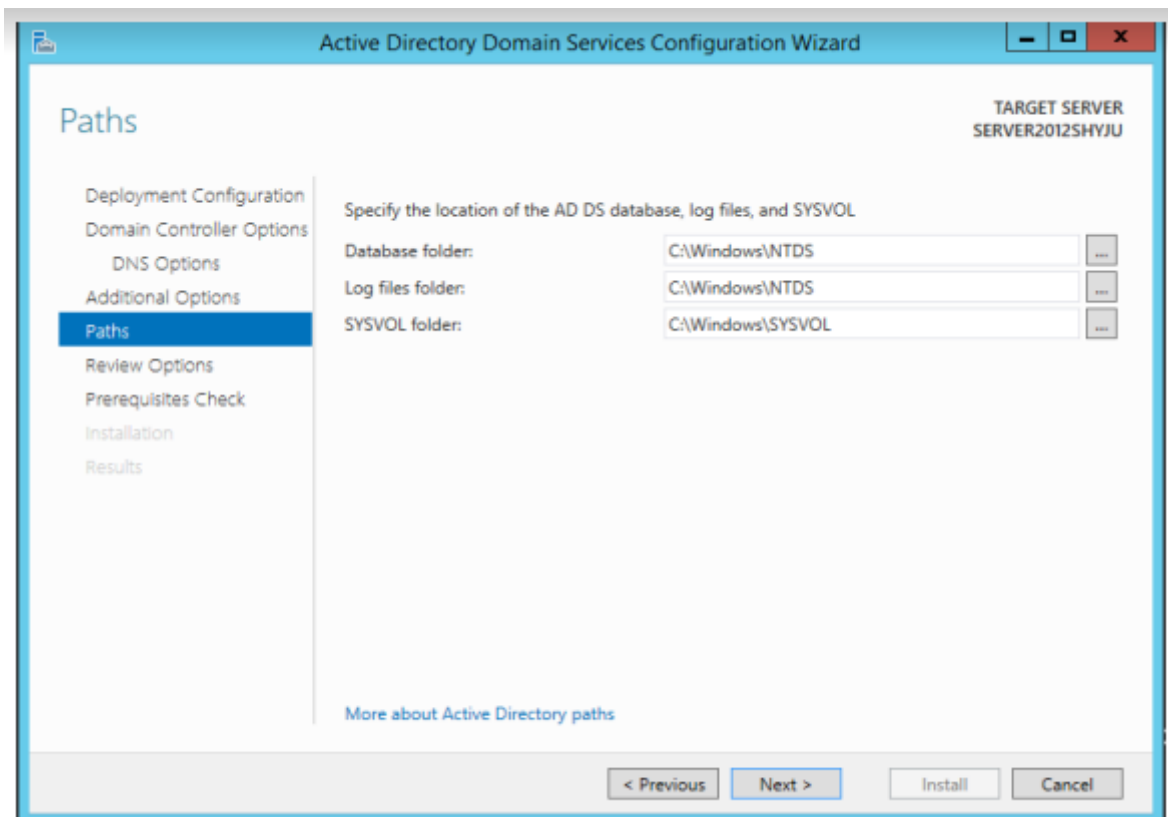


The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window at the 'DNS Options' step. The title bar is the same as the previous screen. The main window has a blue header with the title 'DNS Options' and the same 'TARGET SERVER' label 'SERVER2012SHYJU'. The navigation pane on the left is the same, but 'DNS Options' is now highlighted. The main content area is titled 'Specify DNS delegation options' and contains a single checkbox labeled 'Create DNS delegation', which is currently unchecked. At the bottom of the main area is a link 'More about DNS delegation'. The footer contains the same four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

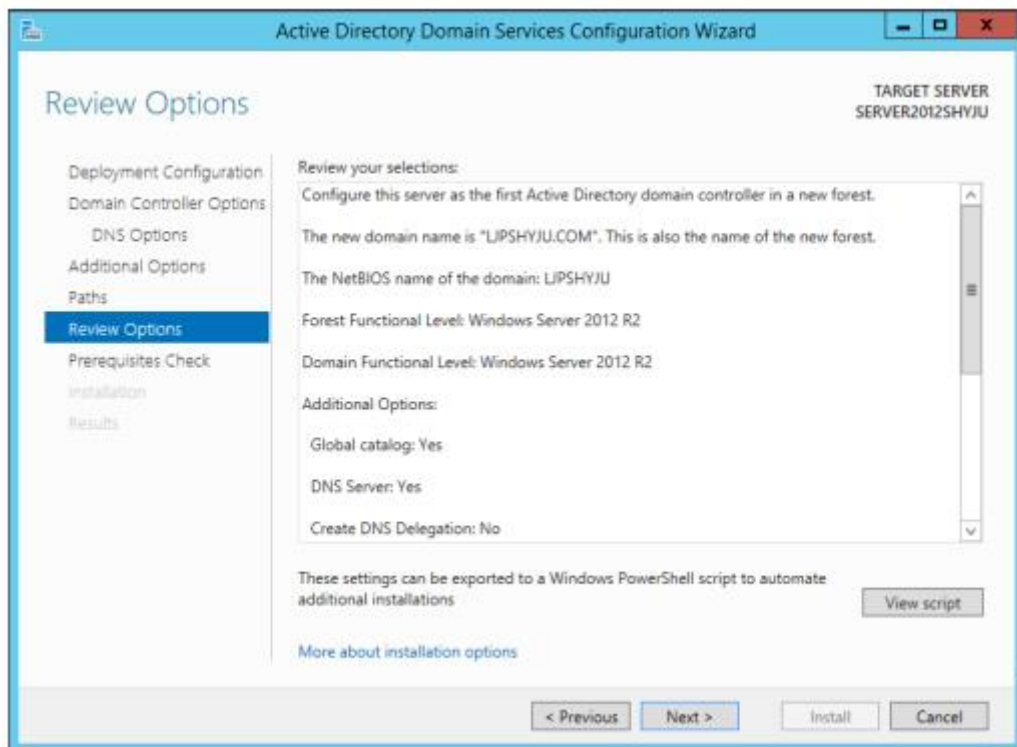
6. Setup will find the NetBIOS domain name. Once found, click on Next.



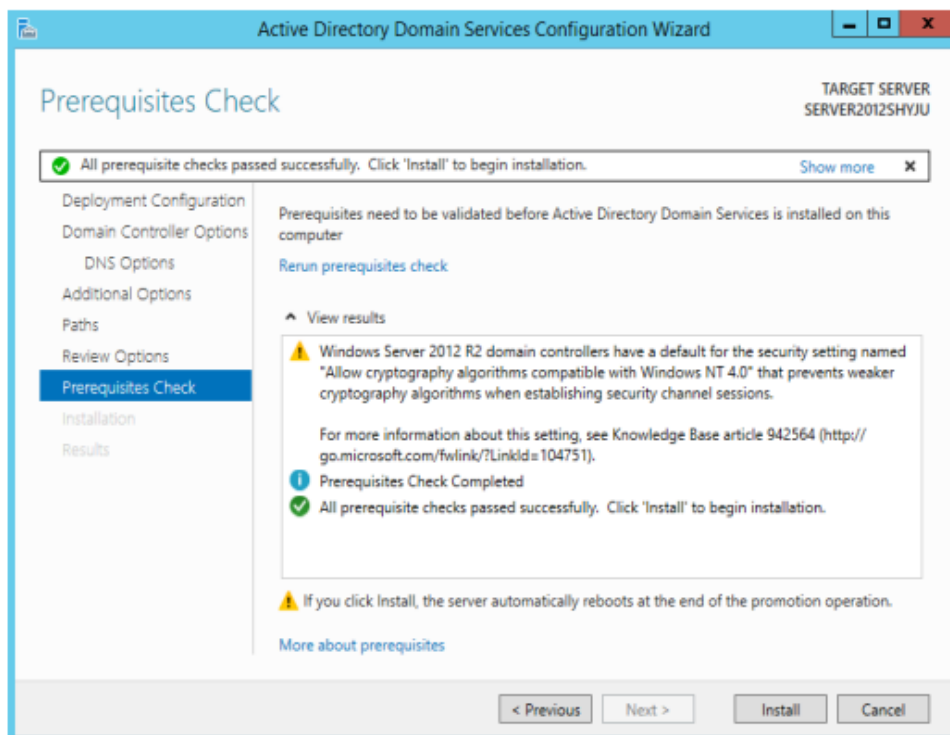
7. Confirm the location of the database, log files and SYSVOL folders.



- Review the summary and click Next to promote the server to Domain Controller.

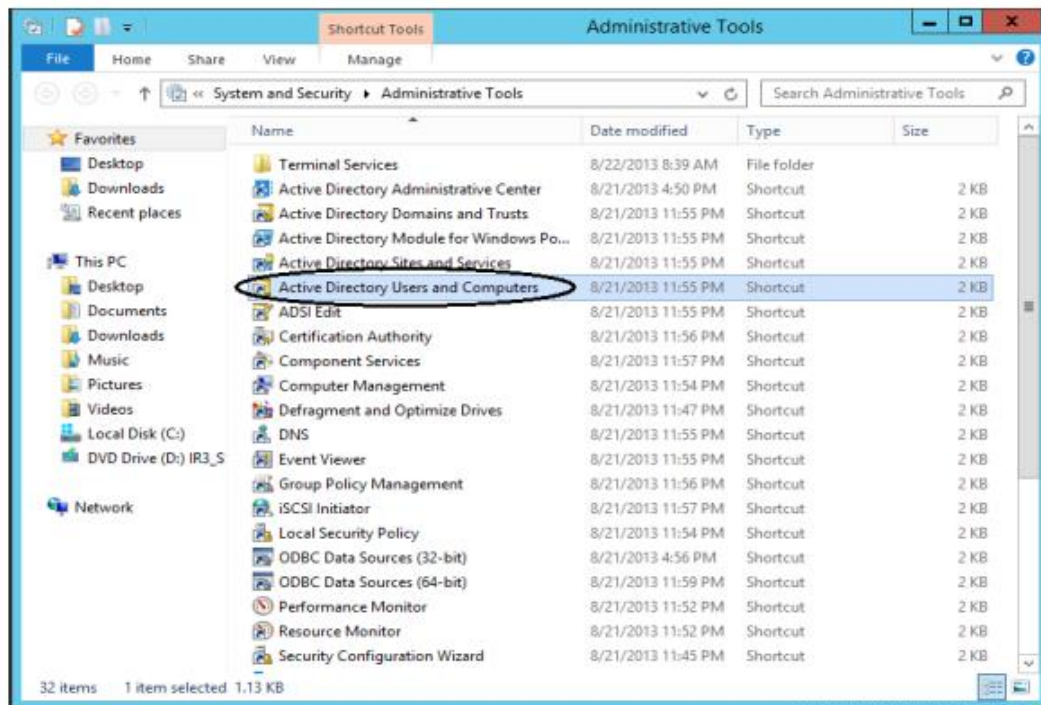


- Once prerequisites have passed successfully, click on Install.



- The system may restart once the installation is completed. Be patient! When the machine restarts and comes back up, you will have an Active Directory Domain controller. To verify if Active Directory tools have been installed, click on the start button, and the Administrative Tools tile. Check out your

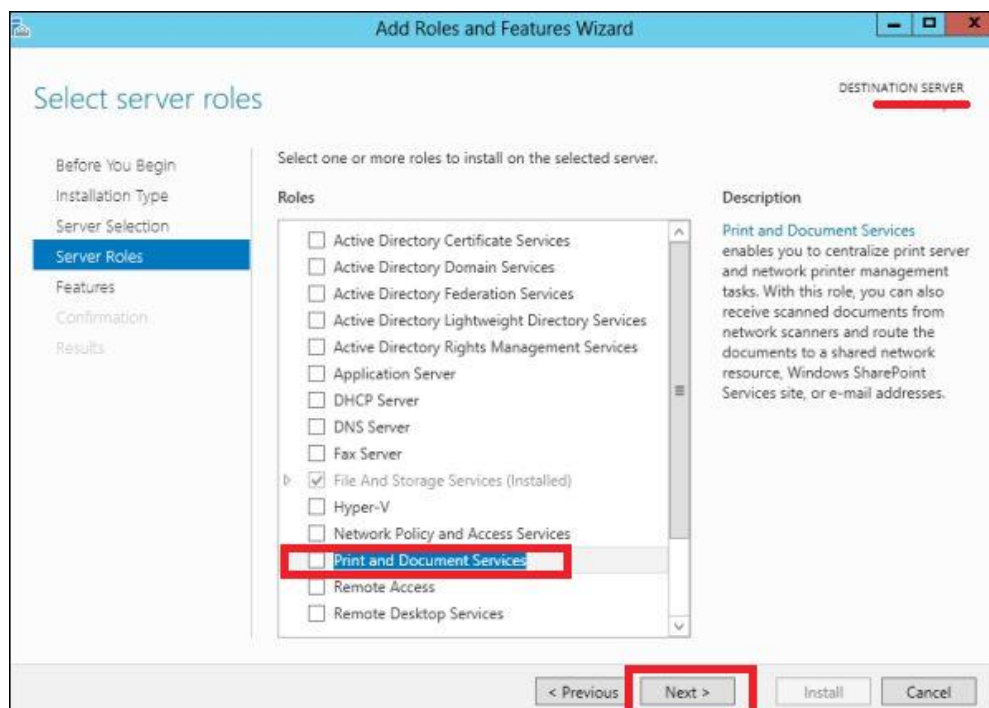
Active Directory tools! The main snap-in for managing users and computers is the Active Directory Users and Computers snap-in.



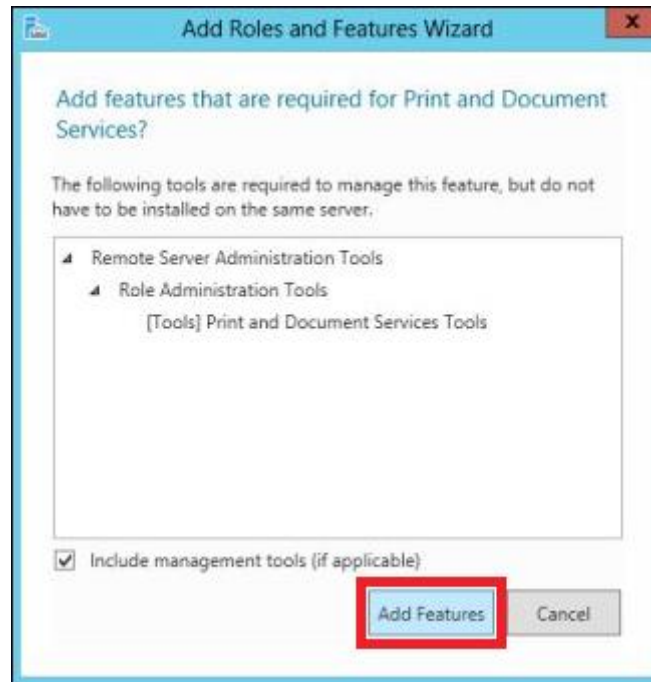
5.3 ADDING FILE SERVER AND PRINTER SERVER

Printer Server

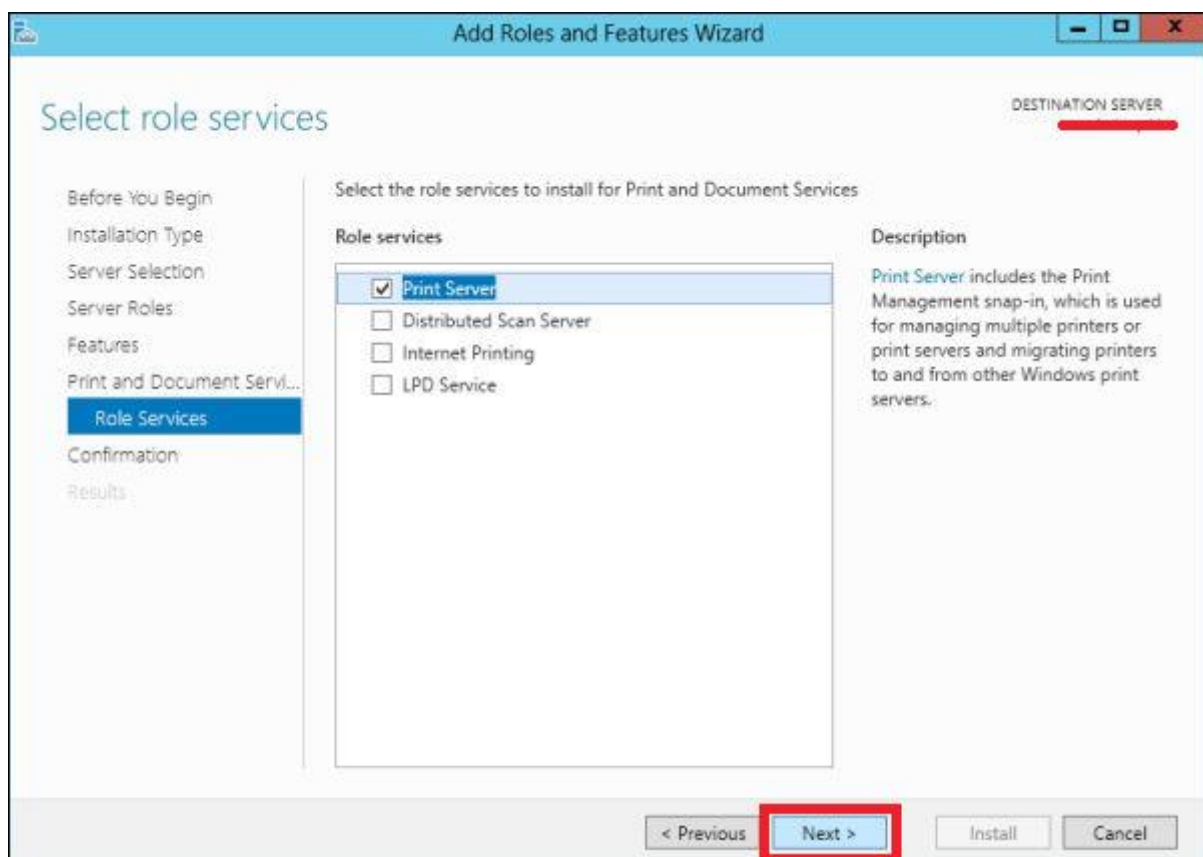
1. Go to Server Manager → Manage → Add Roles and Features → Next → Check on the Role-based or feature-based installation box and → Check on Select a server from the server pool and then → Next. Once it is done then, at the list of the roles find Print and Document Services → A popup window will open.



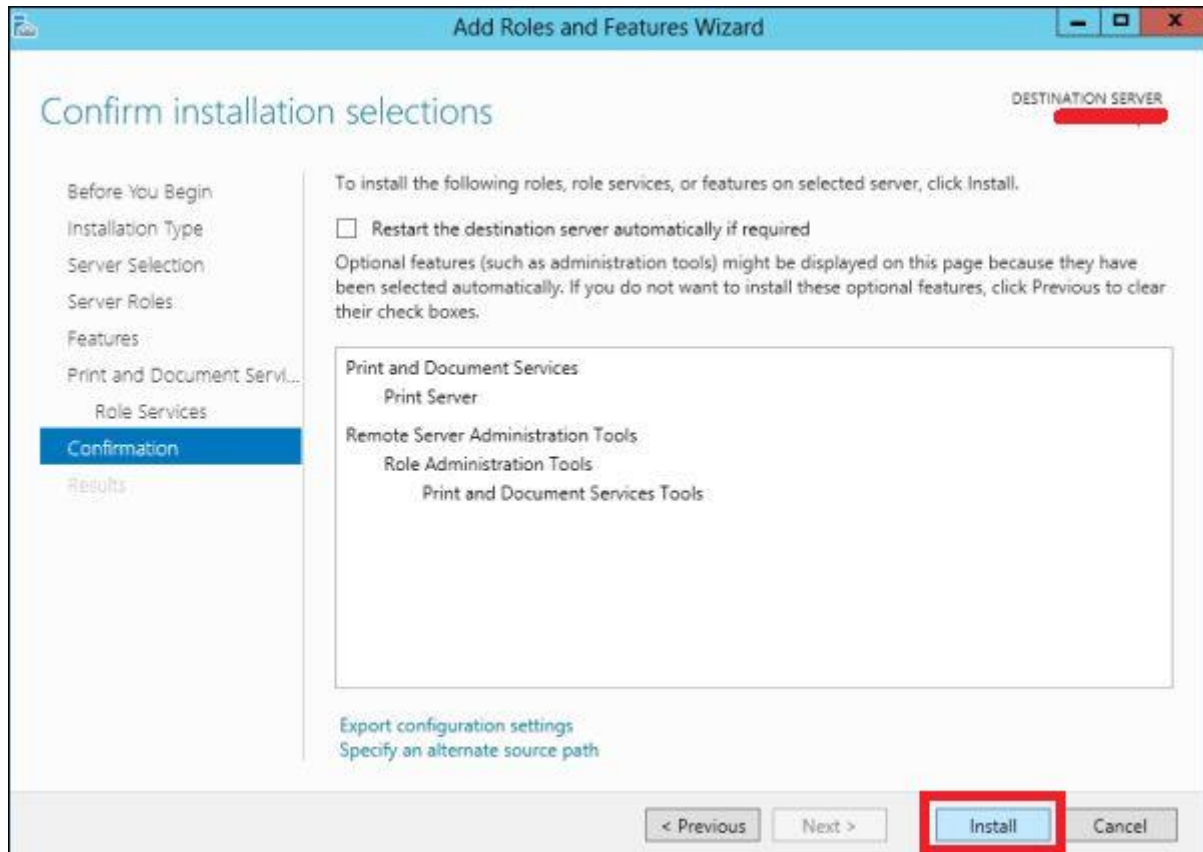
- Click on Add features and then → Next → Next → Next.



- Check on the Print Server box and → Next.

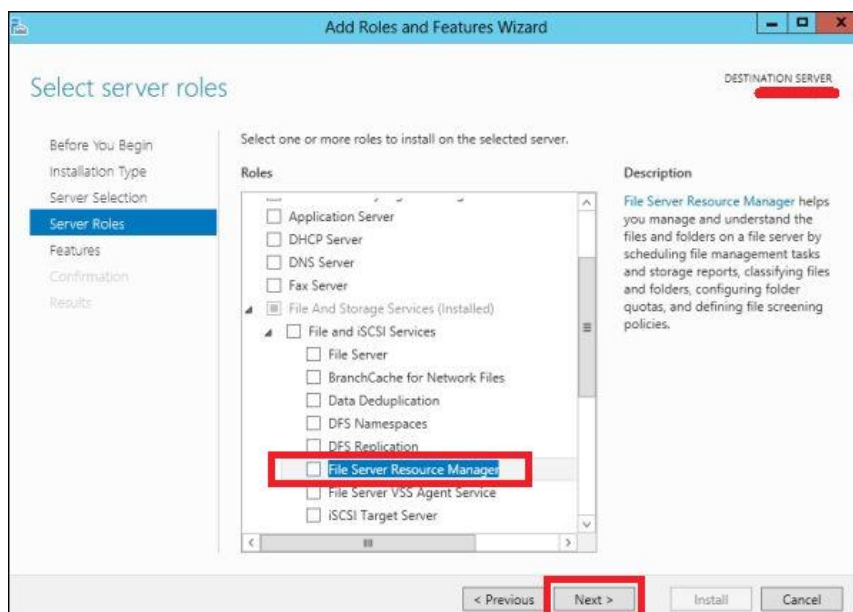


4. Click Install.

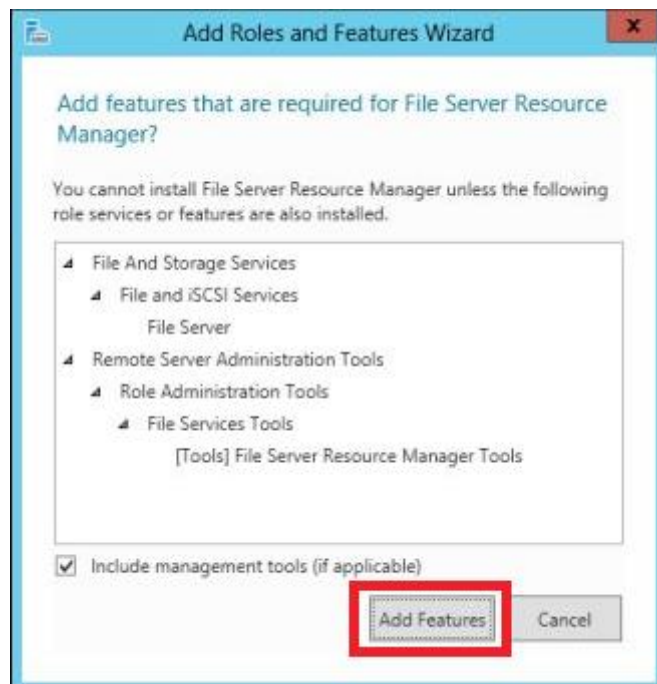


File Server

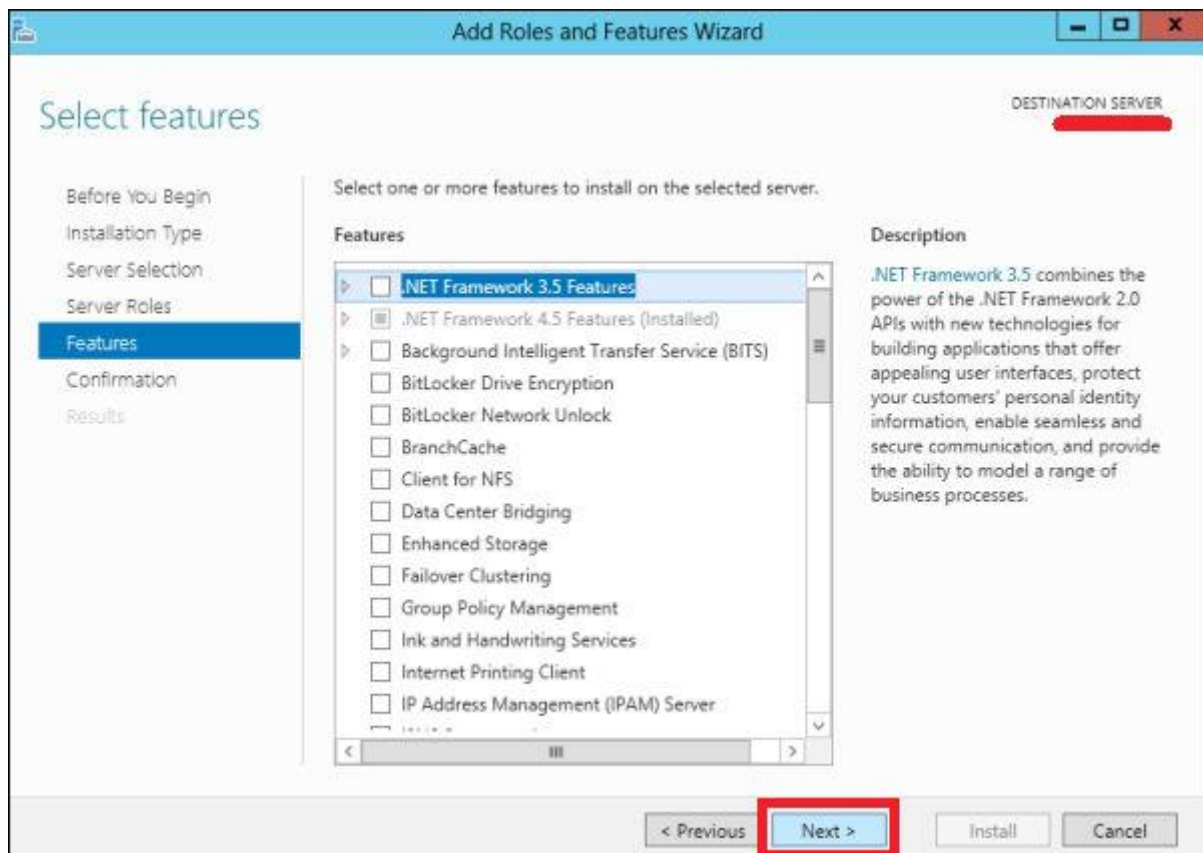
1. Go to Server Manager → Manage → Add Roles and Features → Next → Check on the Role-based or feature-based installation box → Check on the Select a server from the server pool box → Next. Then, at the list of the roles find File and Storage Services and expand it. After this, expand the Files and iSCSI Services → Check on the File Server Resource Manager box and then a pop-up window will open.



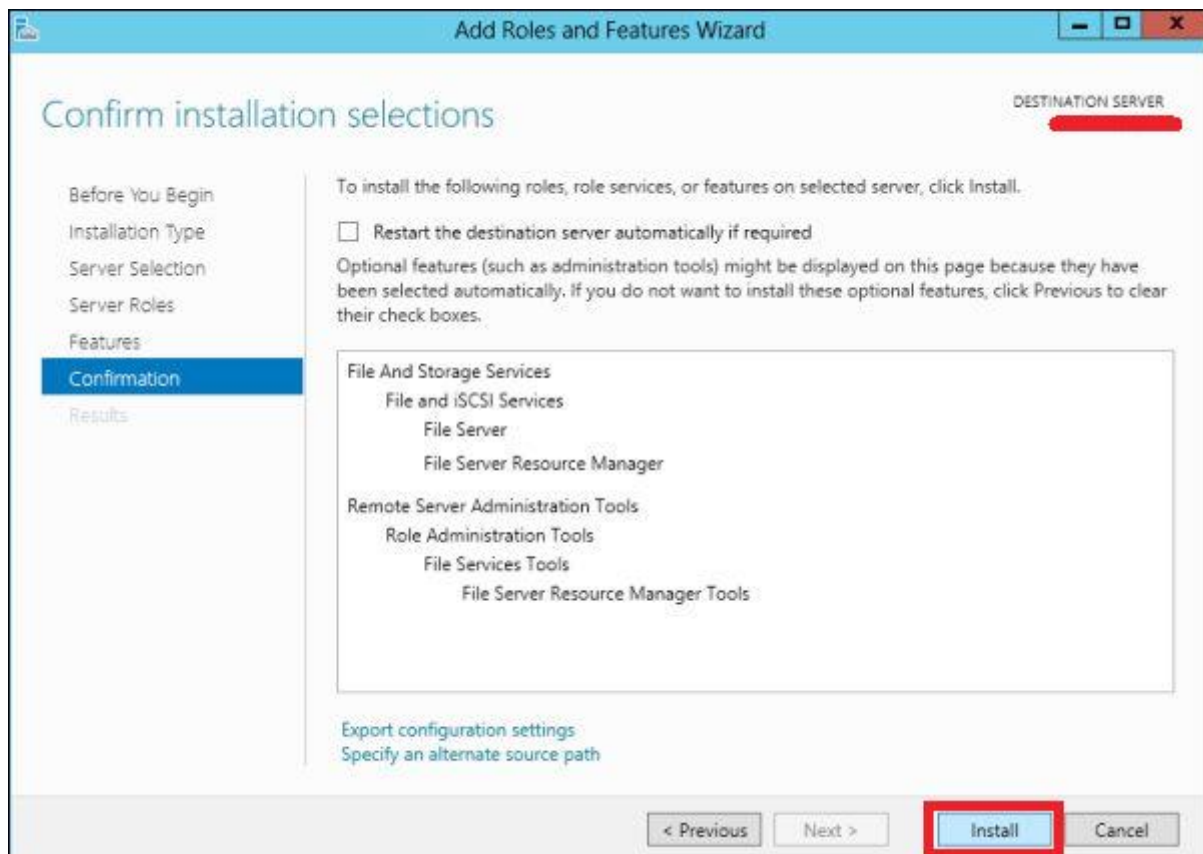
- Click on Add features and then → Next.



- Click the Next button.

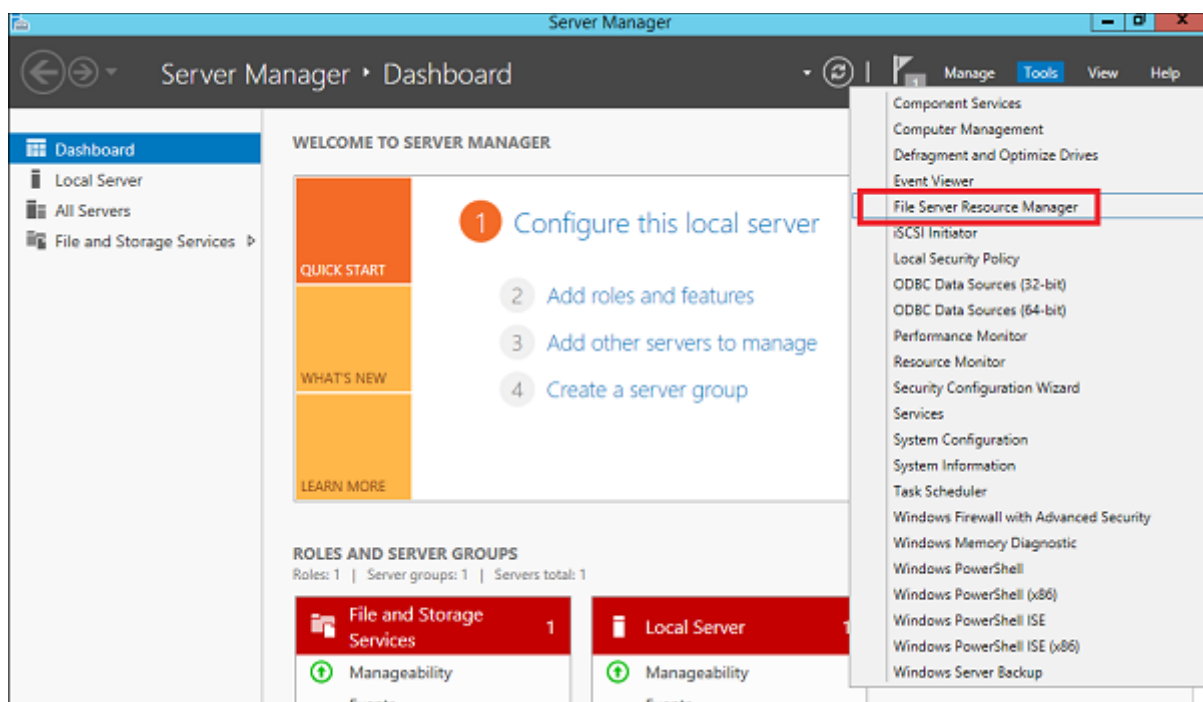


- Click the Install button.



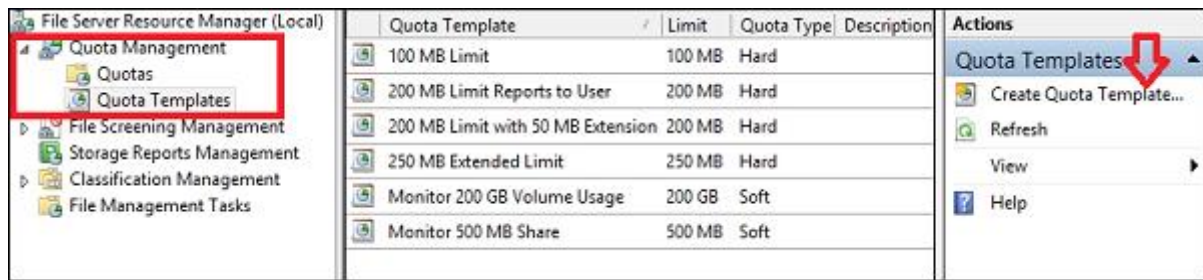
Now that we installed the File Server Resource Manager, let us open it by following the path given below.

- Click on Server Manager → Tools → File Server Resource Manager.



To set quota to a folder, we should follow the steps as below.

- On the left panel click on Quota Management → expand Quota Templates → click on Create Quota Template... on the right-hand side panel as shown in the screenshot given below.



- A new table will be opened, where the most important aspect to put is Space Limit depending on the need. Here, we will put 2GB and then → OK.

The 'Create Quota Template' dialog box is shown. It has a 'Copy properties from quota template (optional):' section at the top with a dropdown menu set to '100 MB Limit' and a 'Copy' button. Below this is the 'Settings' tab. The 'Template name:' field contains '2 GB'. The 'Description (optional):' field also contains '2 GB'. The 'Space limit' section is highlighted with a red box; it shows 'Limit: 2 GB' and the 'Hard quota: Do not allow users to exceed limit' radio button is selected. Below this is the 'Notification thresholds' section with a table for thresholds. At the bottom, the 'OK' button is highlighted with a red box.

Threshold	E-mail	Event Log	Command	Report

- You must put the threshold for it and once the folder reaches the capacity, it will send you a notification where you will have an option to put the email.

Add Threshold

Generate notifications when usage reaches (%):

E-mail Message | Event Log | Command | Report

☐ Send e-mail to the following administrators:

 Format: account@domain. Use semicolons to separate accounts.

☐ Send e-mail to the user who exceeded the threshold

E-mail message

Type the text to use for the Subject line and message.
 To identify the quota, limit, usage, or other information about the current threshold, you can use Insert Variable to insert a variable in your text.

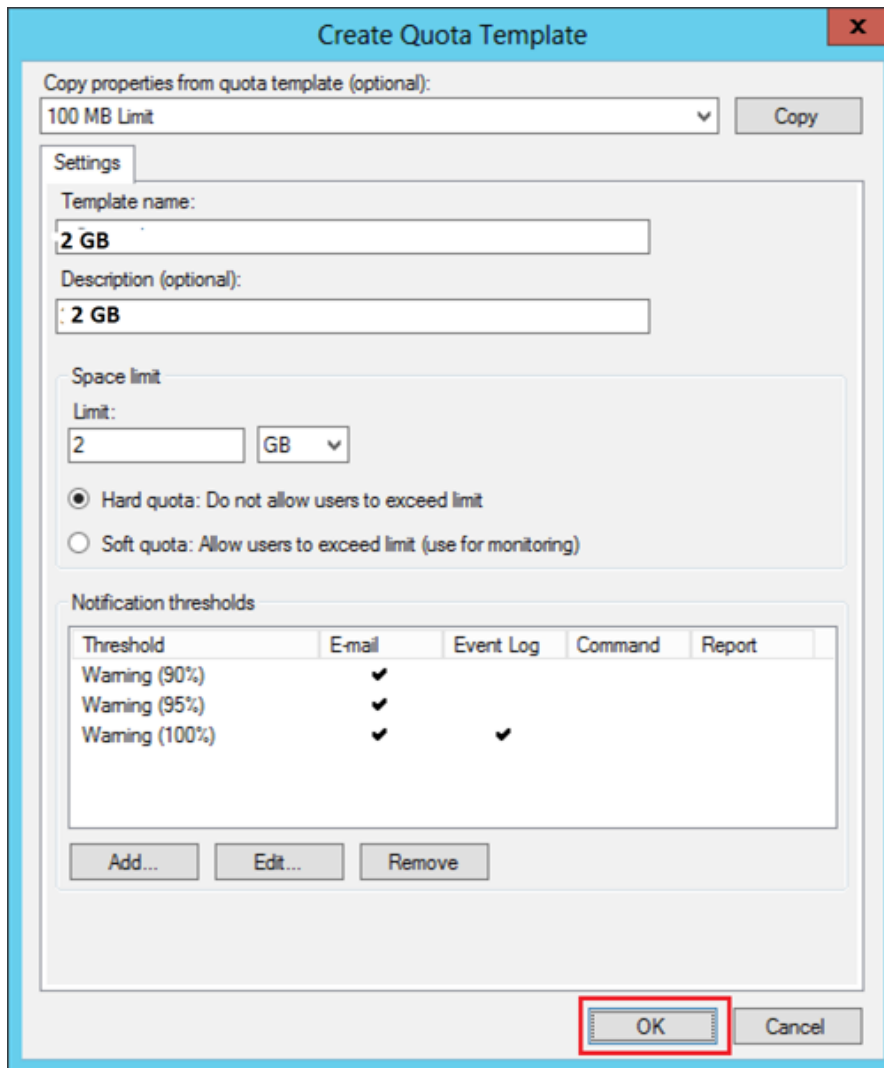
Subject:

Message body:

Select variable to insert:

Inserts the e-mail addresses of the administrators who receive the e-mail.

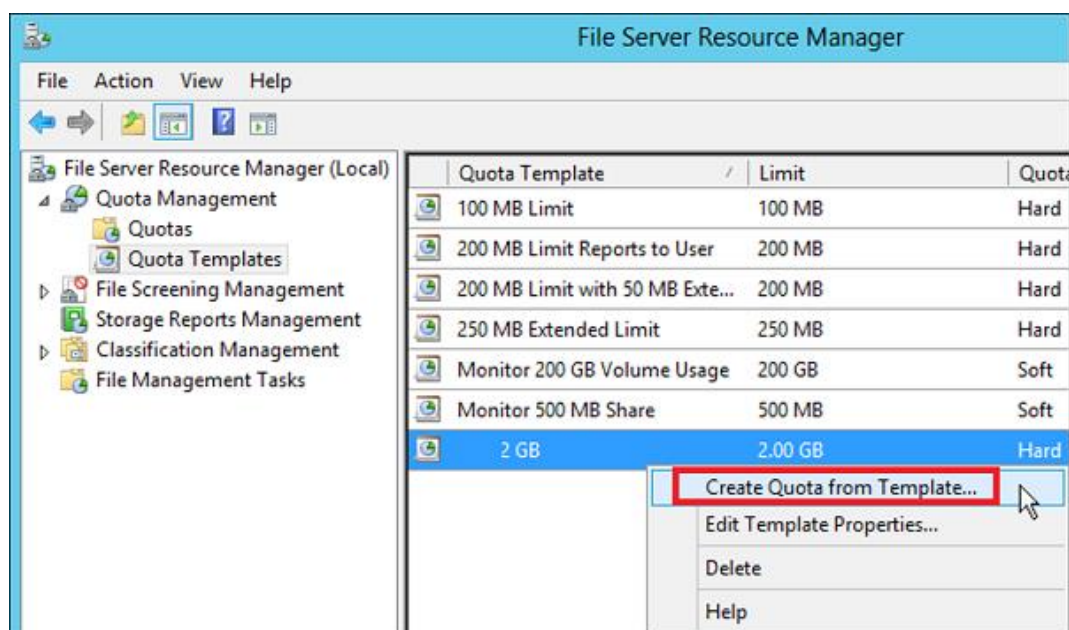
5. Click OK.



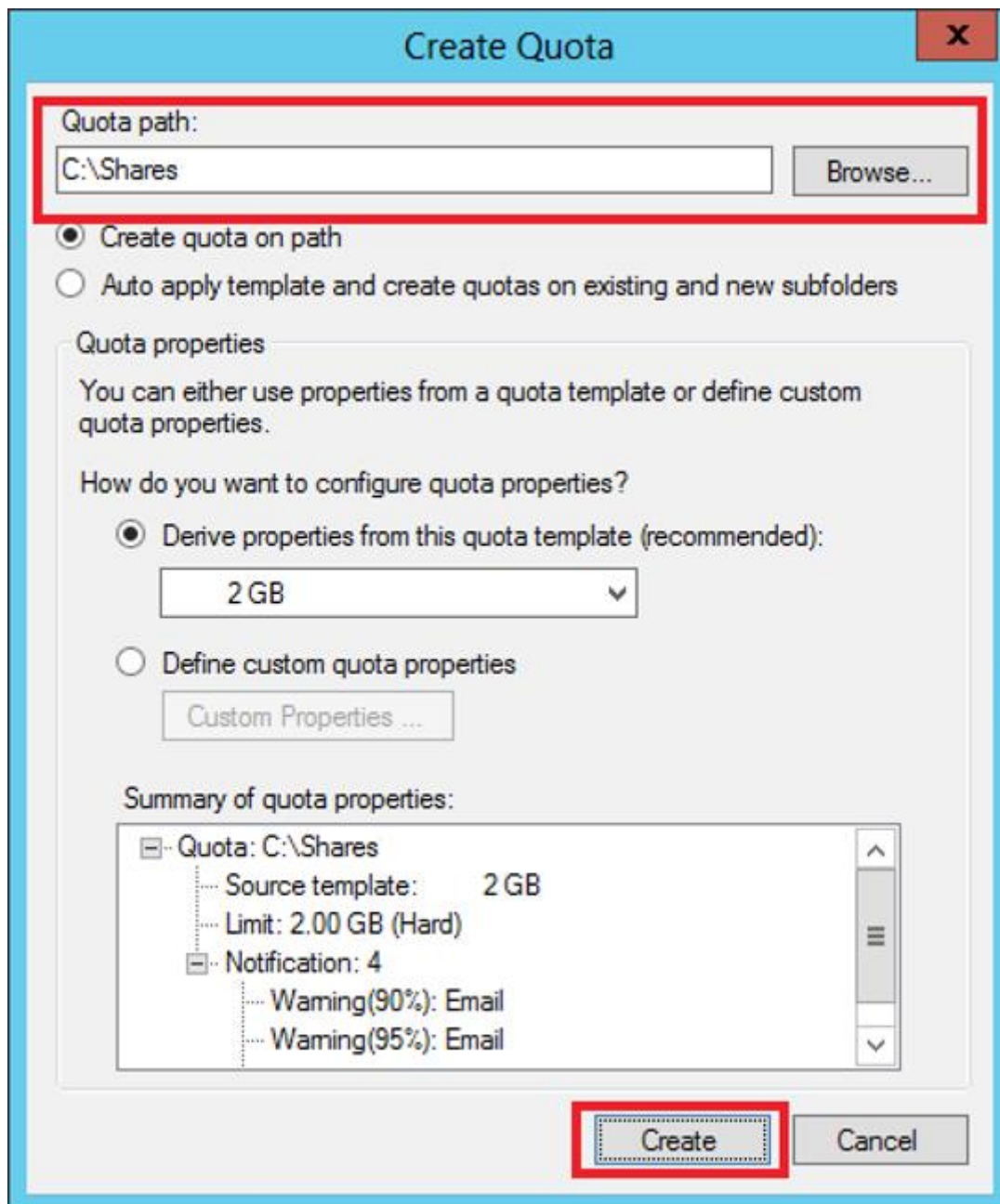
The 'Create Quota Template' dialog box is shown. It has a title bar with a close button. Below the title bar, there is a section 'Copy properties from quota template (optional):' with a dropdown menu set to '100 MB Limit' and a 'Copy' button. The 'Settings' tab is active. It contains a 'Template name:' field with '2 GB' entered, a 'Description (optional):' field with '2 GB' entered, and a 'Space limit' section with a 'Limit:' field set to '2' and a unit dropdown set to 'GB'. Below this, there are two radio buttons: 'Hard quota: Do not allow users to exceed limit' (selected) and 'Soft quota: Allow users to exceed limit (use for monitoring)'. At the bottom, there is a 'Notification thresholds' table with columns 'Threshold', 'E-mail', 'Event Log', 'Command', and 'Report'. The table has three rows: 'Warning (90%)', 'Warning (95%)', and 'Warning (100%)'. The 'E-mail' column has checkmarks for all three rows, and the 'Event Log' column has a checkmark for 'Warning (100%)'. Below the table are 'Add...', 'Edit...', and 'Remove' buttons. At the bottom right, there are 'OK' and 'Cancel' buttons, with the 'OK' button highlighted by a red rectangle.

Threshold	E-mail	Event Log	Command	Report
Warning (90%)	✓			
Warning (95%)	✓			
Warning (100%)	✓	✓		

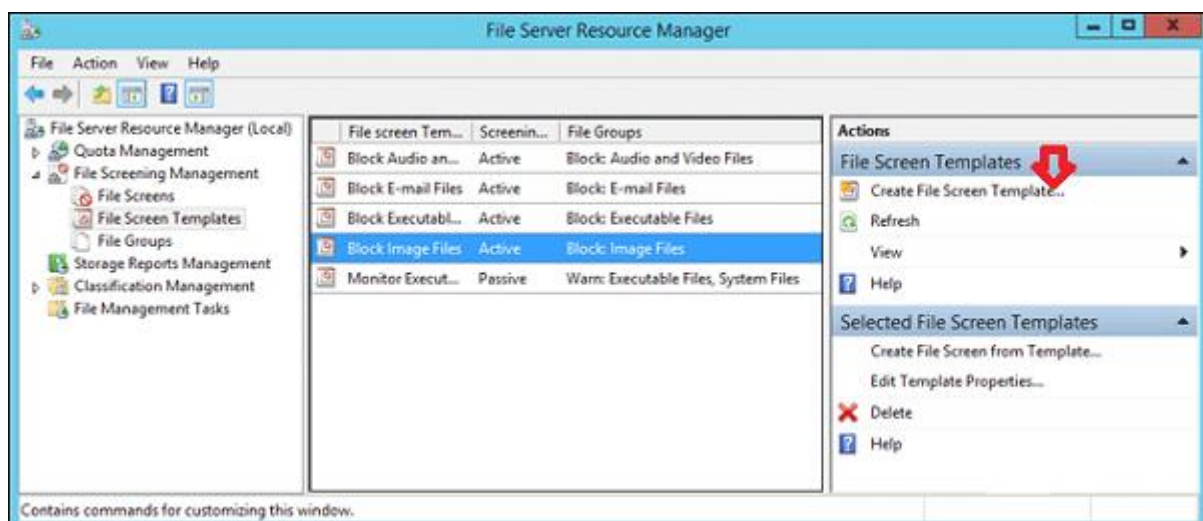
6. Then to attach this quota to a folder – Right Click on the template → click on Create Quota from Template.



7. Click on Browse... and then select your folder → Create.



8. To put a file restriction to your folders, you can go to the left pane File Screening Management → File screening templates → Click on the left panel Create File Screen Template.



- Click on Browse... and find the folder that you want → Create.

Create File Screen

File screen path:

Browse...

File screen properties

You can either use properties from a file screen template or define custom file screen properties.

How do you want to configure file screen properties?

☒ Derive properties from this file screen template (recommended):

Block Image Files

☐ Define custom file screen properties:

Custom Properties ...

Summary of file screen properties:

- File screen:
 - Source template: Block Image Files
 - Screening type: Active
 - File groups: Image Files
 - Notifications: Email, Event log

Create Cancel